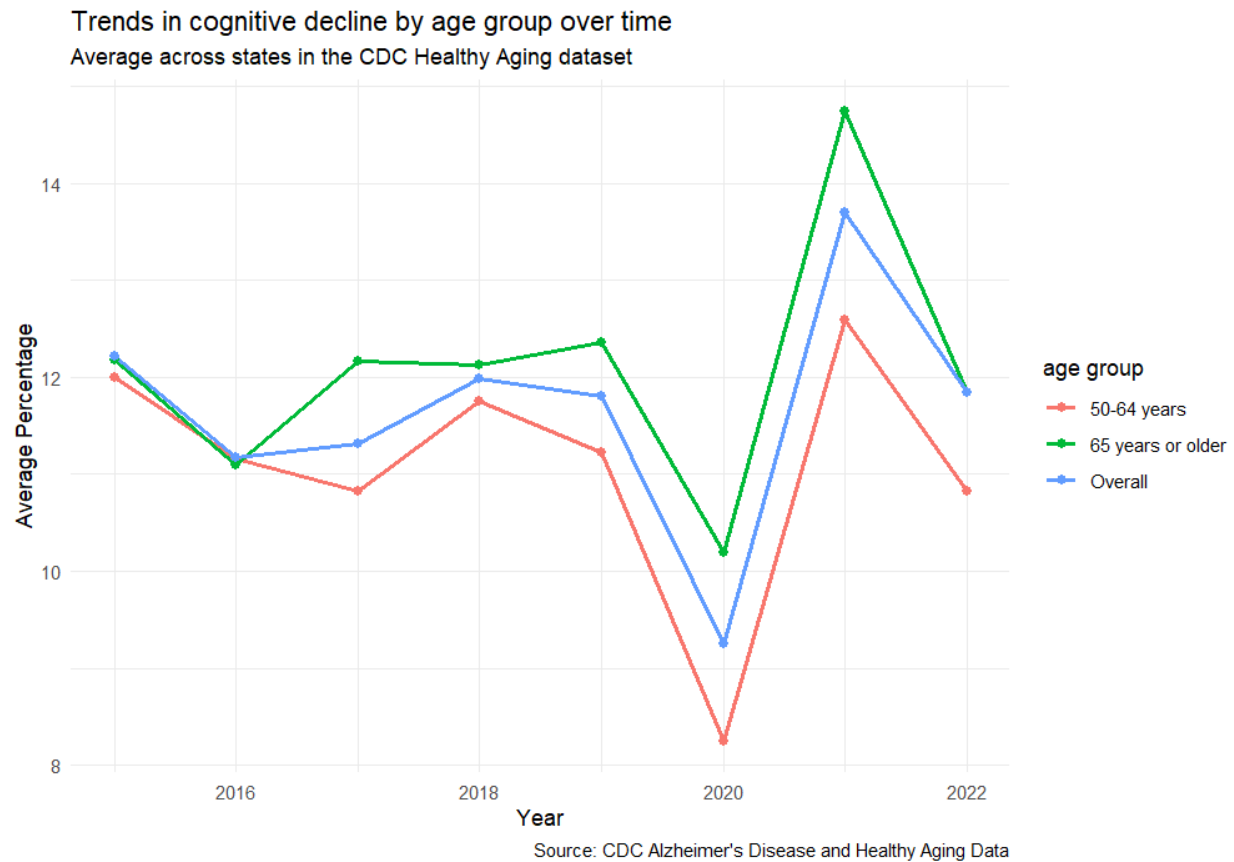
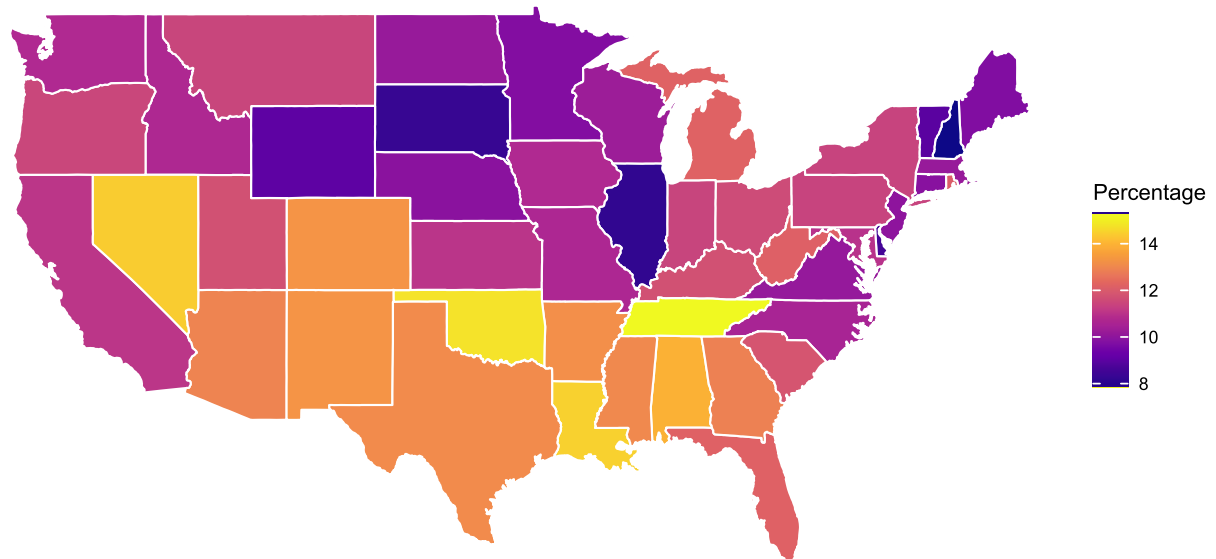




In this graph, I plotted trends in cognitive decline by age group over time using the average across states. I find that the trend is the same across age groups with a dip in 2020 and a spike in 2021, which could be a lack of being able to survey people during Covid pandemic. However, the older the person the higher chance of cognitive decline. I chose to use a line plot because I wanted to visualize data over time and compare the changes across the different age groups.

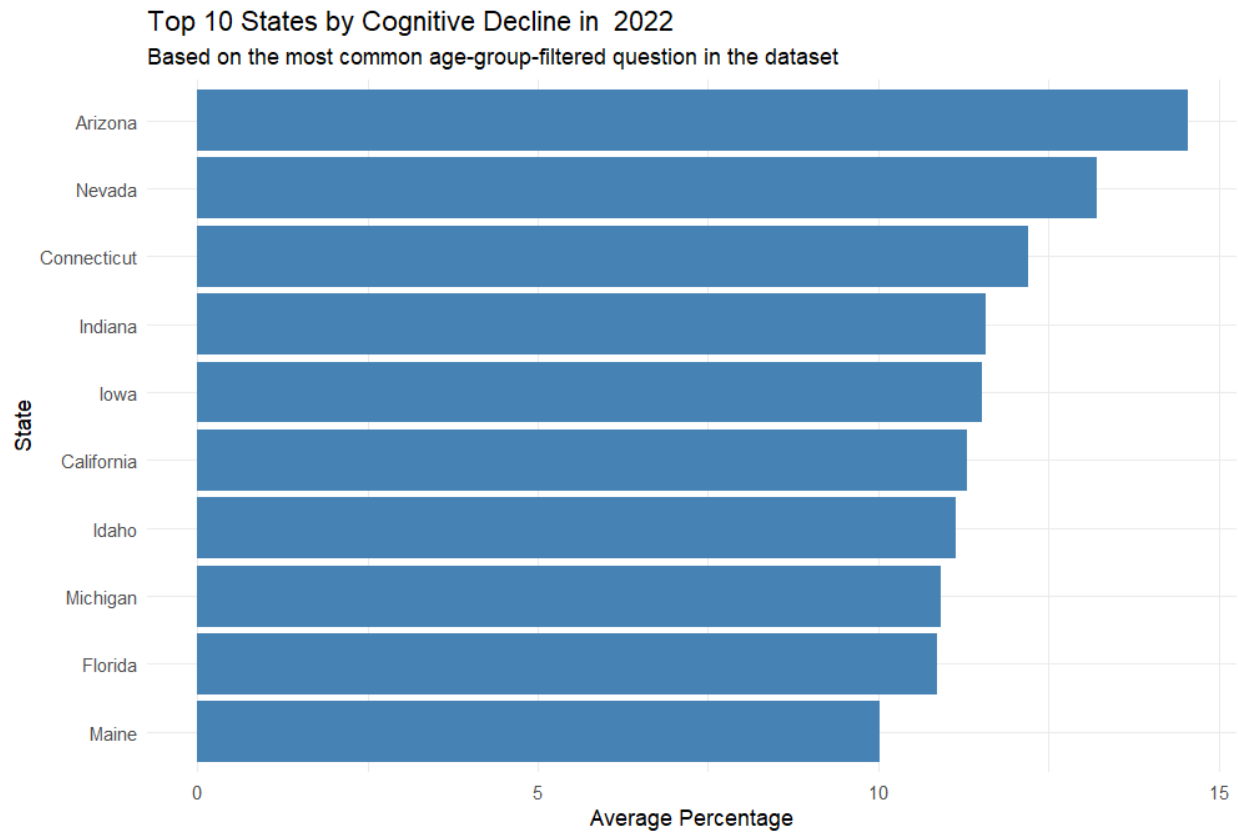
Cognitive Decline Rates by State



Source: CDC Alzheimer's Disease and Healthy Aging Data

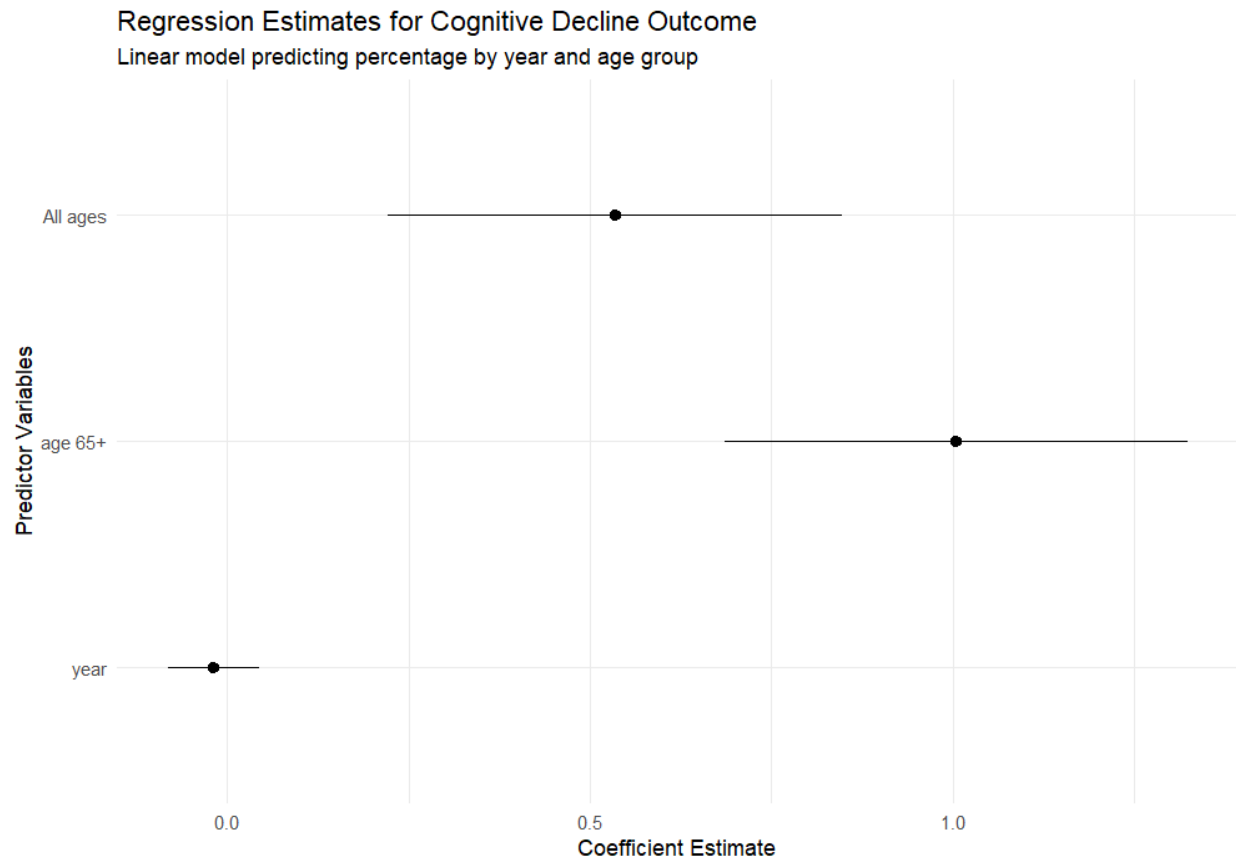
This map shows cognitive decline by state in the US.

States that are more orange and yellow are higher while the darker color shows less cognitive decline. The southern states seem to have higher rates according to this dataset. One reason for this could be due to people moving to the south once they retire, and thus the south has a higher concentration of older people, which correlates to higher cognitive decline. I chose a heatmap because it's a great tool to communicate geographic information and identify places where people have the highest cognitive decline. This could be used as motivation for research into why people in the southern states are more prone to Alzheimer's as basis for research.

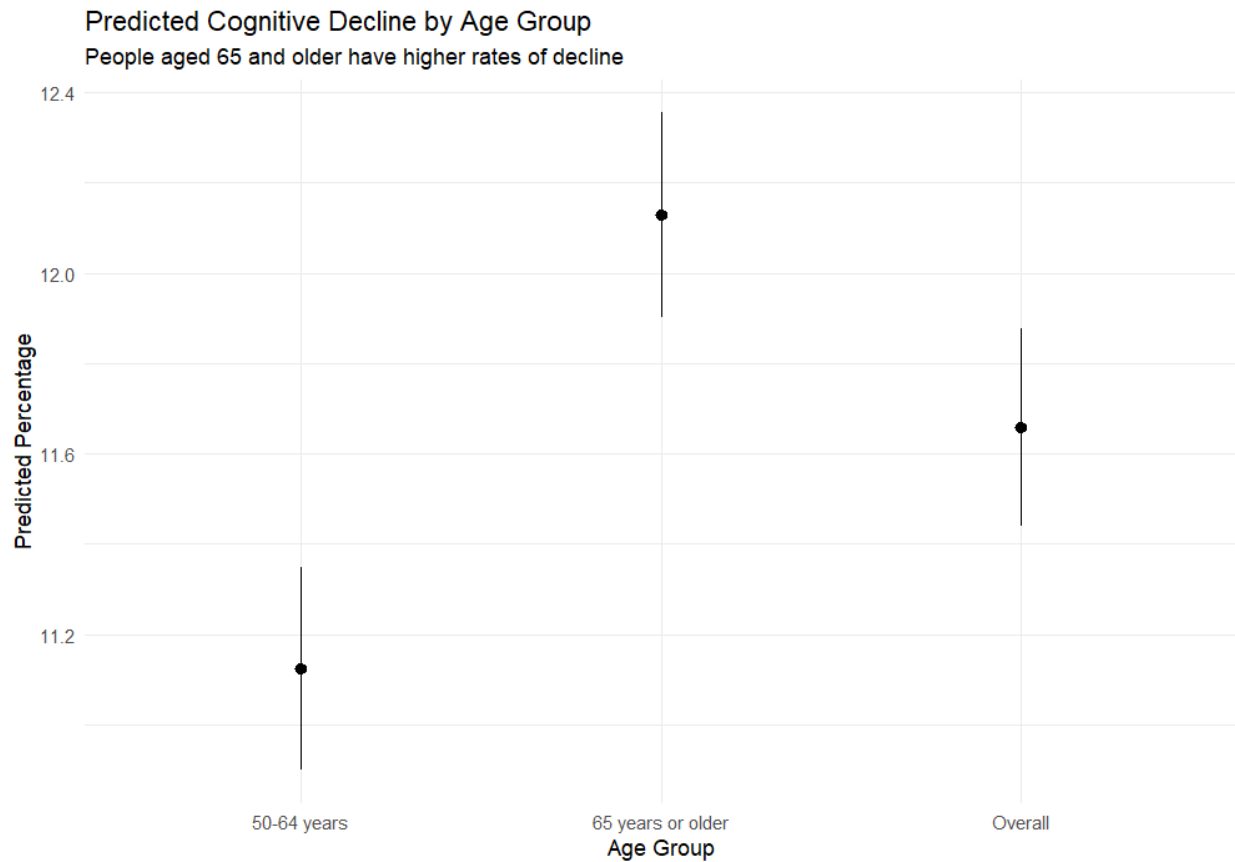


Source: CDC Alzheimer's Disease and Healthy Aging Data

This bar plot shows the top states by cognitive decline in 2022. As of 2022, Arizona has the highest cognitive decline compared to other states. I chose a bar plot to represent this data, because it helped me easily compare rates across the US with most cognitive decline for the year 2022.



I ran a regression on estimates for Cognitive Decline outcome using age and year. This plot shows the estimated effects of age and year on rates of cognitive decline. In terms of time related effects, there is no clear time trend meaning that cognitive decline doesn't change based off of the year. When examining age, there is a large correlation between age and cognitive decline, especially if one is aged 65+. I chose this plot because it clearly shows the coefficients and confidence intervals. It lets me see the effect of age and time and how big that effect is.



Because age played such a role on cognitive decline, I further broke it down by age group. This graph shows predicted cognitive decline depending on age. As we see, the age group of 65+ has a predicted decline of around 12.2 %. On the other hand, people in age group 50 – 64 have a cognitive decline of 11.1 %. I chose this plot because it shows the gap of cognitive decline between these two age groups.